

An Autonomous Recurrence Obstruction for Finite Shape Orbits

Holdout Failure Versus Exact Polynomial Interpolation

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Abstract

The physical quartet shape has one finite monotone coordinate and an exact anisotropic coefficient geometry. It is tempting to infer a scale-independent recurrence. We test that inference on the equal-log-step dyadic subsequence $\sigma = 0.04, 0.02, 0.01, 0.005, 0.0025, 0.00125$.

A two-dimensional affine map fitted separately to the first three transitions interpolates its training data to 10^{-16} , yet its maximum error on the final two transitions is 0.0426 on the left and 0.0506 on the right. A pooled two-channel affine map has training error 0.00205 and holdout error 0.04685. Among scalar axial recurrences, a power-gap transform is best on both channels, but its holdout RMS error remains about 0.0159.

We then prove an exact non-identifiability theorem: every finite two-dimensional orbit whose current axial coordinates are distinct is an orbit of an autonomous polynomial map. Lagrange interpolation constructs $F(u, \eta) = (P(u), Q(u))$ with $F(x_k) = x_{k+1}$. The physical six-state orbit therefore admits a degree-four exact map, with residual below 2×10^{-15} , regardless of whether an underlying semigroup exists.

The combined verdict is negative and structural. The tested low-complexity autonomous maps do not extrapolate reliably, while unrestricted finite recurrence existence has no identifying power. No scale semigroup or continuous generator is obtained.

1 The stronger claim behind a shape clock

RH-214 establishes a finite monotone axial ordering, and RH-217 shows that the coefficient map increasingly suppresses transverse motion [1, 2]. Neither statement implies that equal increments of log scale are generated by one map. That stronger hypothesis would be

$$x_{n+1} = F(x_n), qquad x_n = (u_n, \eta_n), \tag{1}$$

where F is independent of n , channel, and scale.

An autonomous map becomes useful for Gate A only if it has more than interpolation value. It should extrapolate, remain stable as data are added, respect both channels, and ideally embed into a semigroup F_t with $F_{t+s} = F_t \circ F_s$.

2 Equal-log-step subsequence

To avoid confusing unequal scale increments with non-autonomy, we use the dyadic levels

$$0.04, 0.02, 0.01, 0.005, 0.0025, 0.00125. \quad (2)$$

They have constant time step $\Delta t = \log 2$ for $t = \log(1/\sigma)$. The branch coordinates are:

σ	u_L	η_L	u_R	η_R
0.04	0.02027	-0.57806	0.02077	-0.59358
0.02	0.24887	-0.06886	0.24794	-0.07031
0.01	0.40529	-0.10443	0.40229	-0.10748
0.005	0.54098	-0.08934	0.54034	-0.09080
0.0025	0.65874	-0.09050	0.65801	-0.09181
0.00125	0.71835	-0.09303	0.71750	-0.09444

The first three transitions are used for training and the last two for holdout scoring. This split is deliberately harsh: it asks a map fitted through the transient to predict the mature corridor.

3 Scalar axial recurrences

We test three transforms g with

$$g(u_{n+1}) = a g(u_n) + b : \quad (3)$$

$$g_{\text{aff}}(u) = u, qquad g_{\log}(u) = \log \frac{u}{1-u}, \quad g_{\text{gap}}(u) = \log(1-u). \quad (4)$$

model	left		right	
	train RMS	holdout RMS	train RMS	holdout RMS
affine	0.00797	0.01644	0.00926	0.01596
logistic	0.03126	0.11603	0.03226	0.11550
power gap	0.00802	0.01585	0.00942	0.01589

Power gap wins narrowly on both holdouts, consistent with RH-215, but the error is not smaller than the observed fine-scale increments. The logistic recurrence fitted across the transient is decisively poor.

4 Two-dimensional affine audit

Let

$$F(x) = Ax + b, qquad A \in \mathbb{R}^{2 \times 2}, \quad b \in \mathbb{R}^2. \quad (5)$$

There are six scalar parameters. Three two-dimensional transitions provide exactly six scalar equations in generic position, so channelwise training interpolation should not be mistaken for evidence.

On the left, the fitted matrix and offset are approximately

$$A_L = \begin{pmatrix} 0.8505 & -0.0746 \\ 0.0731 & -0.1027 \end{pmatrix}, \quad b_L = \begin{pmatrix} 0.1885 \\ -0.1297 \end{pmatrix}. \quad (6)$$

Its training residual is 7.9×10^{-17} , while its two holdout state errors have maximum 0.04256. On the right, the maximum is 0.05058.

A pooled fit uses six training transitions and one common map:

$$A_{LR} = \begin{pmatrix} 0.8618 & -0.0793 \\ 0.0773 & -0.1044 \end{pmatrix}, \quad b_{LR} = \begin{pmatrix} 0.1843 \\ -0.1324 \end{pmatrix}. \quad (7)$$

Its maximum training error is 0.00205, but its maximum holdout error is 0.04685.

Proposition 4.1 (Finite affine failure). *The tested equal-step affine maps do not provide stable out-of-sample transport of the physical shape orbit: every channelwise or pooled holdout maximum exceeds 0.04.*

Proof. This is the frozen least-squares audit above. The proposition concerns only the declared affine class and split. $\square$

5 Why exact finite recurrence is non-identifying

The opposite danger is to enlarge the model class until interpolation is guaranteed.

Theorem 5.1 (Finite-orbit interpolation). *Let*

$$x_k = (u_k, \eta_k) \in \mathbb{R}^2, \text{quad} k = 0, \dots, n, \quad (8)$$

and suppose $u_0, \dots, u_{n-1}$ are distinct. Then there exists an autonomous polynomial map $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ of the form

$$F(u, \eta) = (P(u), Q(u)) \quad (9)$$

with degrees at most $n - 1$ such that

$$F(x_k) = x_{k+1}, \quad k = 0, \dots, n - 1. \quad (10)$$

Proof. Let $L_k(u)$ be the Lagrange basis polynomials for the distinct nodes $u_0, \dots, u_{n-1}$. Define

$$P(u) = \sum_{k=0}^{n-1} u_{k+1} L_k(u), \quad Q(u) = \sum_{k=0}^{n-1} \eta_{k+1} L_k(u). \quad (11)$$

Then $P(u_k) = u_{k+1}$ and $Q(u_k) = \eta_{k+1}$. $\square$

Corollary 5.2. *The mere existence of an autonomous map fitting a finite monotone shape orbit provides no evidence for a physical semigroup.*

For each six-state physical channel, Theorem 5.1 gives a degree-four map fitting all five transitions. The numerical construction has maximum residual below 2×10^{-15} .

6 What would identify a semigroup

A meaningful positive result would require constraints not supplied by finite interpolation, for example:

1. a map fixed before additional scales are computed;
2. a uniform one-step error tending to zero;
3. consistency across both channels and alternative discretizations;
4. a composition estimate comparing one double step with two single steps;
5. compactness or regularity sufficient to extract a limiting map;
6. for a continuous generator, embeddability into F_t .

The current data satisfy none of these at theorem level. Exact polynomial interpolation actually underscores why such additional structure is needed.

7 Route decision and claim boundary

Further high-degree recurrence fitting to the same fixed quartet is deprioritized. The next question is structural: even if one perfectly controlled this four-root orbit, could it generate the growing zero count required of an infinite determinant? RH-219 answers that question.

This paper identifies no scale semigroup, continuous generator, Fredholm operator, or Gate-A closure. Gates B–E and all zeta/RH statements remain untouched.

References

- [1] Liang Wang. A finite monotone axial clock for the quartet shape flow, 2026. RH-214 preprint.
- [2] Liang Wang. Transverse quenching in quartet coefficient space, 2026. RH-217 preprint.